

17. Answer: D

Given that the distance from the centre of the pattern to the centre of first dark fringe is 1.2 mm, hence the fringe spacing is 2.4 mm.

$$\text{Using } y = \frac{\lambda D}{a}$$

$$\Rightarrow 2.4 \text{ mm} = \frac{450 \times 10^{-9} \times D}{a} \text{ ----- (1)}$$

$$y = \frac{600 \times 10^{-9} \times D}{a} \text{ ----- (2)}$$

$$\frac{(2)}{(1)} \frac{y}{2.4 \text{ mm}} = \frac{600 \times 10^{-9}}{450 \times 10^{-9}}$$
$$y = 3.2 \text{ mm}$$